

RESEARCH ARTICLE

Van der Waals Hybrid-Gate Reconfigurable Transistor Integrating Floating-Gate Logic and Ferroelectric Synaptic Functions

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ABSTRACT

Reconfigurable field-effect transistors (R-FETs) enable multifunctional hardware in which a single device performs several roles. Conventional R-FETs, however, rely on a single gating mechanism, which forces a trade-off between nonvolatile logic reconfiguration and analog synaptic operation. Here, we report a hybrid-gate R-FET that combines floating-gate (FG) and ferroelectric (FE) operation within a single van der Waals gate stack of MoTe₂, *h*-BN, and In₂Se₃. In the FG mode, stored charge strongly dopes the ambipolar MoTe₂ channel, giving stable reconfiguration among insulating, metallic, n-FET, and p-FET states. On this basis, the device reconfigures into inverter and switch operation and, through circuit-level simulation, into six Boolean logic gates within one layout. In the FE mode, polarization switching modulates the channel conductance at low energy, yielding 6-bit synaptic states over a dynamic range of about 100. A hardware-level spiking neural network built on the experimentally measured synaptic characteristics reaches about 90% accuracy on the MNIST dataset in a MATLAB-based simulation. The hybrid-gate R-FET thus provides a versatile device platform that integrates two distinct gating mechanisms and selectively exploits their respective advantages for reconfigurable logic and neuromorphic computing.

1 | Introduction

As the scaling of conventional silicon transistors approaches fundamental physical limits, maximizing circuit functionality within a limited chip area while minimizing power consumption has become an increasingly important challenge [1]. Multifunctional

reconfigurable field-effect transistors (R-FETs) [2–5], capable of integrating various functionalities, including n-/p-FET operation [6, 7], non-volatile memory [8, 9], and synaptic function [10, 11] within a single device have emerged as a promising solution to this challenge. Such integration represents a promising approach toward area- and energy-efficient electronics by reducing the

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number of devices required for a given circuit function. By switching between n- and p-FET modes, R-FETs enable diverse logic operations within a single circuit layout through complementary mode switching [12]. When memory functionality is additionally incorporated, the configured device state can be retained without continuous external bias, thereby significantly reducing the static power consumption of the circuit [13]. Furthermore, by supporting multi-level conductance-state storage in response to external stimuli, R-FETs can emulate hardware-level synaptic functions, thereby extending circuit functionality beyond Boolean logic toward neuromorphic computing [14]. To realize such multifunctional operation, the channel material needs to support electrically controllable carrier polarity while remaining compatible with non-volatile memory layers capable of providing programmable memory functionality. However, conventional CMOS technologies allow only fixed unipolar n- or p-type operation, set by ion implantation, with no runtime switching of conductivity type or carrier polarity [15]. Therefore, electrically reconfigurable device architectures based on electrostatic gate control are essential for realizing dynamic reconfigurability.

In this regard, two-dimensional (2D) van der Waals (vdW) materials serve as an ideal platform for realizing multifunctional R-FETs. Owing to their atomically thin body thickness, 2D vdW materials exhibit exceptionally strong gate electrostatic coupling. This coupling provides excellent gate controllability with suppressed short-channel effects and allows dynamic modulation of both conductivity and carrier polarity through gate-voltage control [16, 17]. In addition, their dangling-bond-free surfaces allow the formation of diverse heterostructures without lattice-mismatch constraints [18]. Building on these advantages, recent studies have actively explored 2D vdW material-based multifunctional R-FETs supporting n-/p-FET switching, memory, and synaptic operations. These R-FETs are typically realized through two major non-volatile device architectures, (i) floating-gate [19–24] and (ii) ferroelectric-gate [25–27] structures. The floating-gate architecture enables stable and large-capacity charge storage, as the floating gate is electrically isolated between two insulating layers (tunneling and blocking layers). This configuration allows wide-range modulation of channel conductivity and clear polarity switching through gate-bias control. However, the tunneling-based write/erase operation (e.g., Fowler–Nordheim tunneling) typically requires program/erase voltages exceeding several tens of volts and relatively slow write/erase operation [28]. These factors increase the energy consumption per weight update and limit suitability for synaptic applications that demand frequent, low-energy, and fast conductance updates. In contrast, ferroelectric-gate architectures operate at low voltages with fast switching speeds through electric-field-induced polarization switching, making them well suited for power-efficient synaptic operation [29]. However, since the ferroelectric polarization-driven conductivity modulation typically provides a narrower memory window compared to floating-gate architectures owing to the limited remnant polarization-induced charge. This narrower window makes it challenging to achieve the wide and reversible conductivity modulation required for reliable n-/p-FET switching. Therefore, to overcome this trade-off, a device architecture that can selectively utilize both mechanisms is required. In such an architecture, floating-gate operation provides wide-range conductivity modulation for n-/p-FET switching,

while ferroelectric-gate operation enables low-power, high-speed synaptic weight updates.

Here, we demonstrate a multifunctional R-FET that incorporates floating-gate and ferroelectric non-volatile mechanisms within a single 2D vdW gate stack, hereafter referred to as a “hybrid-gate R-FET”. This device operates as a floating-gate memory under high gate voltage and as a ferroelectric memory under low gate voltage. This dual operation allows the dominant mechanism to be selectively activated depending on the functional requirements of the target application. The floating-gate mechanism is utilized for reconfigurable logic, which demands strong electrostatic doping but does not require frequent memory updates. In this mode, the device supports metallic, insulating, n-FET, and p-FET states, with each transistor exhibiting an on/off current ratio exceeding 10^5 , enabling dynamic logic reconfiguration. Based on this capability, we experimentally demonstrate dynamic switching between inverter and switch configurations. Circuit-level simulation further establishes that a single circuit layout can support six Boolean gates, including XOR, OR, AND, XNOR, NOR, and NAND. In contrast, the ferroelectric mechanism is applied to synaptic operations, where memory updates occur frequently and thus low operating voltage and fast update speed are essential. In this mode, the device exhibits a dynamic range of $\sim 10^2$ in response to pulse voltages incremented from 2.5 to 10 V, enabling effective synaptic weight modulation. Based on the experimentally extracted synaptic characteristics, hardware-level simulations of a spiking neural network (SNN) achieve MNIST digit recognition accuracy exceeding $\sim 90\%$. This hybrid-gate approach addresses the inherent trade-off between the two major R-FET architectures, providing a unified device platform for multifunctional, energy-efficient electronics that integrate reconfigurable logic with neuromorphic computing.

2 | Results and Discussion

2.1 | Dual-Mode Non-Volatile Operation of the Hybrid-Gate R-FET

Figure 1a illustrates the device structure of the proposed hybrid-gate R-FET, which integrates two distinct non-volatile mechanisms within a single gate stack. To realize this hybrid-gate concept, we designed a $\text{MoTe}_2/\text{h-BN}/\text{In}_2\text{Se}_3$ van der Waals heterostructure on a SiO_2/Si substrate with Pt/Ag source/drain contacts. Here, the ambipolar MoTe_2 serves as the channel layer, where its ambipolar carrier transport enables effective modulation of both n-type and p-type conduction for reconfigurable logic [30] and its excellent carrier mobility supports reliable synaptic transistor operation [31]. The h-BN layer functions as a tunneling barrier for charge injection, and In_2Se_3 acts as the storage layer that plays a dual role in the hybrid-gate operation. Because the bandgap of In_2Se_3 is positioned between those of the adjacent wide-bandgap dielectrics SiO_2 and h-BN (Figure S1), it forms a charge-storage well. Under a high gate electric field, charge carriers injected via Fowler–Nordheim tunneling are stored in this well, enabling floating-gate operation [32]. At the same time, the intrinsic ferroelectricity of In_2Se_3 [33] allows the device to operate as a ferroelectric memory under a low electric field. The operating concept of the floating-gate (FG) mode is schematically illustrated in Figure 1b. In the FG

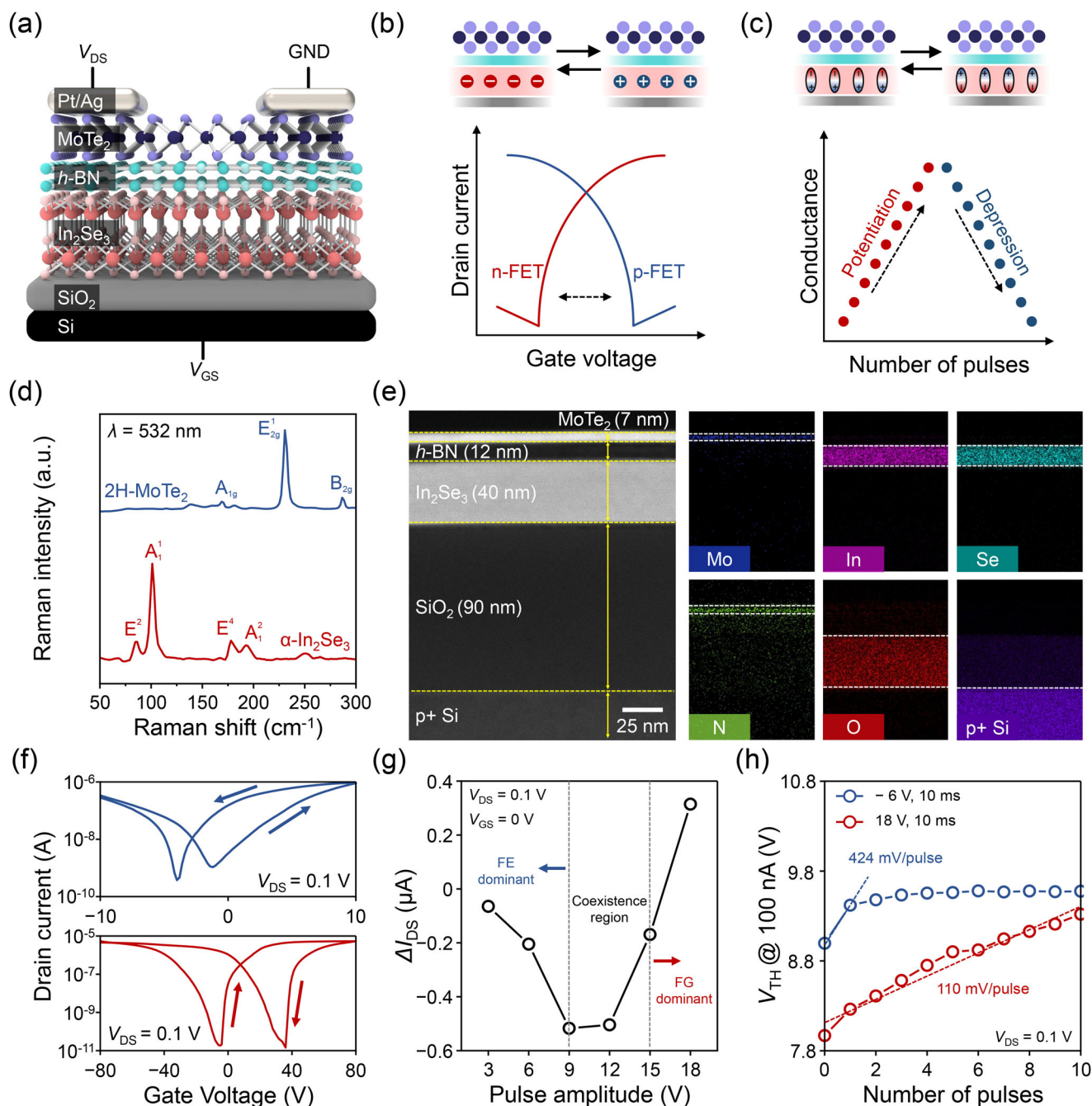


FIGURE 1 | Device structure and hybrid-gate operation of the R-FET. (a) Schematic of the proposed hybrid-gate R-FET based on a MoTe₂/h-BN/In₂Se₃ van der Waals heterostructure. (b,c) Schematic illustrations of the target application of the hybrid-gate R-FET: (b) FG mode for reconfigurable logic and (c) FE mode for synaptic operation. (d) Raman spectra of the MoTe₂ and In₂Se₃ flakes used in the hybrid-gate R-FET. (e) Cross-sectional TEM image (left) and corresponding EDS elemental mapping (right) of the hybrid-gate R-FET. (f) I_{DS} - V_{GS} hysteresis characteristics measured with different gate-voltage sweep ranges, corresponding to FE mode operation at a low voltage range (top) and FG mode operation at a high voltage range (bottom). (g) Extracted drain-current change (ΔI_{DS}) as a function of programming pulse amplitude, showing the FE dominant region, the FE-FG coexistence region, and the FG dominant region. (h) Evolution of threshold-voltage shift with programming pulse number, showing fast but saturating programming in the FE mode and slow but non-saturating cumulative programming in the FG mode.

mode, charge carriers trapped through the tunneling barrier electrostatically dope the ambipolar MoTe₂ channel with a large doping strength. This shifts the threshold voltage and enables reconfigurable n-FET/p-FET operation, which demands high doping levels without frequent reprogramming. Through the cumulative storage of charge carriers in In₂Se₃, the device polarity can be precisely programmed and maintained for reconfigurable

logic applications. Figure 1c presents the operating concept of the ferroelectric (FE) mode. In this mode, the switchable out-of-plane remnant polarization of the ferroelectric In₂Se₃ layer modulates the channel conductance at a high programming speed. This enables synaptic potentiation and depression characteristics requiring fast, energy-efficient weight updates driven by external stimuli. Unlike the FG mode, the FE mode utilizes

reversible polarization switching to achieve rapid conductance modulation suitable for neuromorphic computing applications.

To identify the crystal phases of the MoTe_2 and In_2Se_3 flakes used in our devices, we conducted Raman spectroscopy, as shown in Figure 1d. MoTe_2 can exist either in the ambipolar semiconducting 2H phase or in the semi-metallic 1T' phase [34]. Because R-FET operation requires gate-controlled modulation of both channel conductance and carrier polarity, the 2H phase is required. In contrast, In_2Se_3 exhibits several polymorphs, including the α -, β -, and γ -phases. To integrate ferroelectric memory functionality, the α -phase is required, as it uniquely possesses out-of-plane ferroelectric polarization originating from its non-centrosymmetric quintuple-layer structure [35]. the α -phase is required, as it uniquely possesses out-of-plane ferroelectric polarization originating from its non-centrosymmetric quintuple-layer structure. The in-plane and out-of-plane polarization components of α - In_2Se_3 originate from the same lateral displacement of the central Se layer and are therefore intrinsically coupled [36]. As shown in Figure 1d, the Raman spectrum of MoTe_2 displays characteristic A_{1g} , E_{2g}^1 , and B_{2g} modes, confirming that the MoTe_2 flakes used in this work are in the 2H phase [37]. Meanwhile, the In_2Se_3 flakes exhibit distinct E^2 , A_1^1 , E^4 , and A_1^2 vibrational modes corresponding to the α -phase [38]. These results confirm that both materials possess the desired crystal phases for dual-mode R-FET operation. In addition to the Raman analysis, cross-sectional transmission electron microscopy (TEM) and energy-dispersive x-ray spectroscopy (EDS) analyses were performed to verify the structural integrity of the fabricated heterostructure. As shown in Figure 1e, the MoTe_2 (7 nm)/ h -BN (12 nm)/ In_2Se_3 (40 nm) stack exhibits clean and well-defined interfaces without noticeable interfacial degradation. Furthermore, the thicknesses of the constituent layers were carefully designed to support efficient charge storage. In particular, the 90-nm SiO_2 layer serves as a blocking dielectric to suppress charge leakage, while the 12-nm h -BN layer functions as a tunneling barrier that enables charge injection into the In_2Se_3 storage layer under a high electric field. The thickness variation and corresponding estimated layer number of In_2Se_3 were further evaluated by AFM measurements on four representative flakes comparable to those used in the devices (Figure S2).

The transfer characteristics of the hybrid-gate R-FET measured at $V_{DS} = 0.1$ V are presented in Figure 1f. Under a narrow gate voltage sweep range (-10 to 10 V), where tunneling is expected to be negligible, the device exhibits a counterclockwise hysteresis loop in the positive gate voltage region. This low-voltage range corresponds to an FE dominant regime, where the upward out-of-plane polarization of In_2Se_3 induces electron accumulation in the channel, thereby shifting the threshold voltage in the negative direction. In contrast, when the gate voltage sweep range is extended to ± 80 V, corresponding to a gate voltage range within the FG dominant regime, the hysteresis reverses to a clockwise loop. This indicates that the net charge storage in In_2Se_3 via tunneling induces hole accumulation in the channel, resulting in a positive shift of the threshold voltage. These contrasting hysteresis behaviors indicate that the dominant memory mechanism changes from FE polarization switching at low gate voltages to FG charge storage at high gate voltages.

To further clarify the distinction between the two operating modes, we analyzed the doping effect induced by gate-voltage pulses with different amplitudes, as shown in Figure 1g. The doping behavior of our device originates from two competing mechanisms. In the FE mechanism, a positive gate pulse aligns the polarization of In_2Se_3 upward, toward the h -BN tunneling barrier, inducing electron accumulation in the channel and resulting in n-type electrostatic doping. In contrast, in the FG mechanism, a sufficiently large positive gate pulse drives carrier exchange across the h -BN tunneling barrier. Holes are emitted from In_2Se_3 into the channel and, at higher gate voltages, electrons are additionally injected from the channel into In_2Se_3 . Both processes increase the net negative charge in the storage layer, which electrostatically induces p-type doping of the channel. Therefore, the FE and FG mechanisms produce doping effects in opposite directions. When both mechanisms coexist, their competing contributions can partially compensate each other, resulting in a reduced net modulation of the channel current [39].

To experimentally examine this competing behavior, the device was initialized to a comparable p-branch current level of approximately 2.5 to 5.2×10^{-6} A at $V_{GS} = 0$ V before each programming condition. The change in drain current (ΔI_{DS}) was then extracted after applying programming pulses with different amplitudes (Figure S3). Four successive stages are identified in Figure 1g, and the corresponding band diagrams are presented in Figure S4. In the low-voltage region from 3 to 9 V, where the contribution of FN tunneling is expected to be negligible, the magnitude of the current decrease grows monotonically with the pulse amplitude, from -0.06 μA at 3 V to -0.52 μA at 9 V, indicating that the out-of-plane polarization of In_2Se_3 evolves from a partially aligned state to a fully aligned state. Between 9 and 12 V, ΔI_{DS} remains essentially unchanged at -0.52 and -0.51 μA , indicating that the polarization is already fully aligned at approximately 9 V and that no further FE contribution is available. At 15 V, the magnitude of ΔI_{DS} decreases to -0.17 μA even though the pulse amplitude continues to increase. Since the FE contribution is already saturated, this reduction is attributed to holes that are emitted from In_2Se_3 into the channel through the h -BN barrier, which is the lower-barrier of the two charge-transfer processes (Figure S1), whose doping effect is opposite to that of the polarization and therefore partially screens it. At 18 V, the sign of ΔI_{DS} is reversed to $+0.32$ μA , indicating that the stored charge fully screens the polarization-induced field and that the charge-storage-induced p-type modulation exceeds the FE-induced n-type modulation. Accordingly, we define the region below 9 V as the FE mode, the region between 9 and 15 V as the FE-FG coexistence region, and the region above 15 V as the FG mode in this work. These results demonstrate that the relative contributions of the FE and FG mechanisms, and thus the dominant non-volatile operating mode, can be controlled by the gate-voltage amplitude.

To further support the onset of the competing FG contribution, the voltage and electric-field distribution across the gate stack was quantitatively estimated using an equivalent capacitance model (see Figure S5). Approximately 11.6% of the applied gate voltage drops across the h -BN tunneling barrier, and a comparison with the band offsets of the stack indicates that hole emission from

In_2Se_3 , which presents the lowest barrier of approximately 1.5 eV in the stack, becomes possible at a gate voltage of approximately 13 V. This value lies within the FE-FG coexistence region identified in Figure 1g.

Figure 1h compares the programming dynamics of the two memory modes by monitoring the threshold voltage shift as a function of the number of applied pulses. Under the FE mode with a -6 V gate pulse, the device exhibits a threshold voltage shift of 424 mV per pulse. This corresponds to a high memory update speed that allows the device state to be rapidly reconfigured with a small number of pulses. In contrast, under the FG mode with a 18 V gate pulse, the shift per pulse is reduced to 110 mV, corresponding to a lower memory update speed. This contrast originates from the fundamentally different physical mechanisms governing each mode. In the FE mode, the polarization of In_2Se_3 switches collectively upon each pulse application, producing a substantial modulation of the channel electric field per programming event. However, because the total switchable polarization is finite, the threshold voltage shift gradually saturates with continuous pulse application. In the FG mode, charge carriers are injected through the h -BN tunneling barrier and accumulate in In_2Se_3 , producing a smaller threshold voltage shift per pulse. The magnitude of the threshold voltage shift, however, remains relatively constant even with prolonged pulse application, indicating that the charges can be cumulatively stored without saturation. These distinct programming characteristics are well-aligned with the requirements of each target application. The high memory update speed of the FE mode supports frequent, energy-efficient weight updates for synaptic operation. The sustained, cumulative doping capability of the FG mode enables precise definition of the device polarity for reconfigurable logic. Together, these complementary characteristics establish the hybrid-gate design strategy proposed in this work.

2.2 | Reconfigurable n-/p-FET in the FG Mode of the Hybrid-Gate R-FET

Figure 2a shows the energy band diagrams of the programming mechanism of the FG mode in the hybrid-gate R-FET. The four configurable states are accessed through two complementary programming sequences, hole programming and electron programming, followed by readout at $V_{\text{GS}} = 0$. Under a sufficiently large negative gate voltage ($V_{\text{GS}} \ll 0$), a strong electric field develops across the h -BN tunneling layer. FN tunneling driven by the electric field simultaneously extracts electrons from In_2Se_3 and injects holes into the storage layer, completing the hole programming (step i). When the bias is subsequently returned to $V_{\text{GS}} = 0$, the trapped positive charges remain stored between the SiO_2 and h -BN barriers. These charges electrostatically induce electron accumulation in the MoTe_2 channel, thereby producing n-type doping of the device (step ii). Conversely, under a sufficiently large positive gate voltage ($V_{\text{GS}} \gg 0$), the band tilt reverses, allowing the trapped holes to be removed and electrons to be injected into In_2Se_3 (step iii). At $V_{\text{GS}} = 0$, the stored negative charges induce hole accumulation in the channel, thereby producing p-type doping (step iv). Together, these two complementary programming sequences enable controllable modulation of the channel doping type with stable retention of

the programmed state, providing the basis for reconfigurable logic operation.

Figure 2b presents the transfer characteristics ($I_{\text{DS}}-V_{\text{GS}}$) measured after applying programming pulses of opposite polarity. A -50 V pulse applied for 1.5 s configures the device into the n-FET state, whereas a $+50$ V pulse applied for 1 s sets the p-FET state. These relatively high programming voltages are required to induce a sufficiently large nonvolatile threshold-voltage shift for switching between the n-FET and p-FET states within the same gate-voltage read range used in our reconfigurable logic scheme. Both charge-transfer processes across the h -BN barrier have to contribute to the stored charge for a full polarity switch, and the process with the higher barrier becomes active only above approximately 24 V under positive bias and 29 V under negative bias (Figure S1), so that ± 50 V provides sufficient margin. The programming duration is set by the amount of charge that has to be stored, since the stored charge first compensates the opposing ferroelectric contribution before a net threshold-voltage shift develops. Programming conditions of several tens of volts have also been reported in previous back-gated reconfigurable devices, as summarized in Table S1. In both polarity configurations, the device exhibits high-performance transistor behavior, with on-currents of 0.5 μA (n-FET) and 5 μA (p-FET) and on/off current ratios exceeding 10^5 . To evaluate the operational reliability of these reconfigured states, we performed endurance and retention measurements (Figure 2c,d). Under 100 repeated hole-electron programming cycles, the device maintains a current ratio between the n-FET and p-FET states above 10^3 at $V_{\text{GS}} = 7$ V. This stability indicates reproducible polarity inversion without progressive degradation. The retention measurement extends this evaluation up to 10^4 s, during which both polarity configurations retain an on/off current ratio above 10^4 . These results show that the FG mode supports stable and reproducible reconfigurable operation. The reproducibility of the hybrid-gate operation was evaluated on three independently fabricated devices. All of them reproduced the reversal of the hysteresis direction between the two operating modes, the reconfiguration between the n-FET and p-FET states, and the endurance and retention characteristics (Figure S6 and Table S2).

Beyond polarity inversion between the n-FET and p-FET states, the strong electrostatic doping capability of the FG mode, combined with the ambipolar transport of the MoTe_2 channel, enables further modulation of the transistor characteristics. In a unipolar channel, a comparable threshold-voltage shift would only translate the transfer curve without changing the conducting carrier type, whereas the ambipolar transport of MoTe_2 makes both the electron and hole branches accessible within the same gate-voltage window. As shown in Figure 2e, this combination enables access to four distinct conductance states within the same gate-voltage window. In the insulating state, the device maintains an off-current of $\sim 10^{-10}$ A across the entire gate window. As the threshold voltage shifts toward the negative (positive) direction, the device exhibits typical n-type (p-type) FET behavior within the same window. With further shift in the same direction, the channel enters a metallic state characterized by a high current of $\sim 10^{-6}$ A that remains nearly constant regardless of the applied gate voltage. Figure 2f presents the switching characteristics among the four conductance states under a sequence of programming pulses applied for 2 s each. The pulse amplitudes follow the

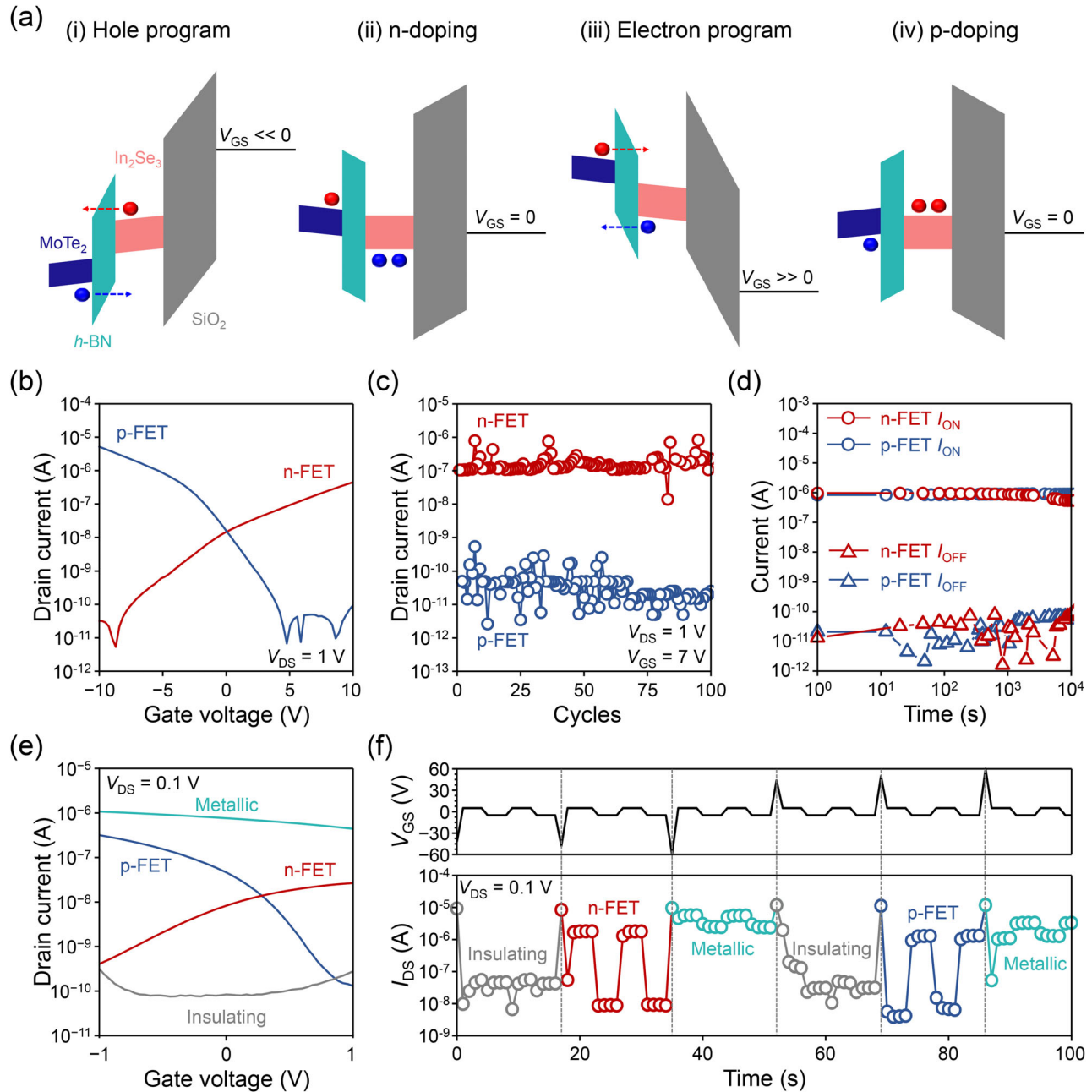


FIGURE 2 | Electrical characteristics of the hybrid-gate R-FET operated in the FG mode. (a) Energy band diagrams illustrating the FG programming mechanism for n-type and p-type doping of the MoTe₂ channel. (b) I_{DS} - V_{GS} transfer characteristics of the programmed n-FET and p-FET states in the FG mode. (c) Cycling endurance of the programmed n-FET and p-FET states, showing reproducible current levels during repeated polarity reconfiguration. (d) Retention characteristics of the programmed n-FET and p-FET states, showing stable ON and OFF currents up to 10^4 s. (e) I_{DS} - V_{GS} transfer characteristics of the four programmable conductance states in the FG mode. (f) Dynamic switching among the four programmable conductance states under sequential programming pulses.

order of -42 , -48 , -60 , $+44$, $+50$, and $+60$ V. Between successive programming pulses, small-amplitude gate pulses are introduced to probe the transfer characteristics and monitor the on/off behavior. The device transitions sequentially through insulating, n-FET, metallic, insulating, p-FET, and metallic states, showing clear and reproducible switching between the four conductance levels. These results demonstrate the high tunability of the R-FET, where the four programmable states distinguish the present work from previous studies that combined the FE and FG effects of In₂Se₃ to enhance synaptic characteristics [32], while providing

enhanced flexibility for the multifunctional reconfigurable logic circuits described in the next section.

2.3 | Reconfigurable Boolean Logic Implemented With the FG-Mode R-FET

Figure 3a illustrates the circuit configuration and device structure of the two-transistor R-FET unit for reconfigurable inverter and switch operation. Here, two hybrid R-FET devices are connected

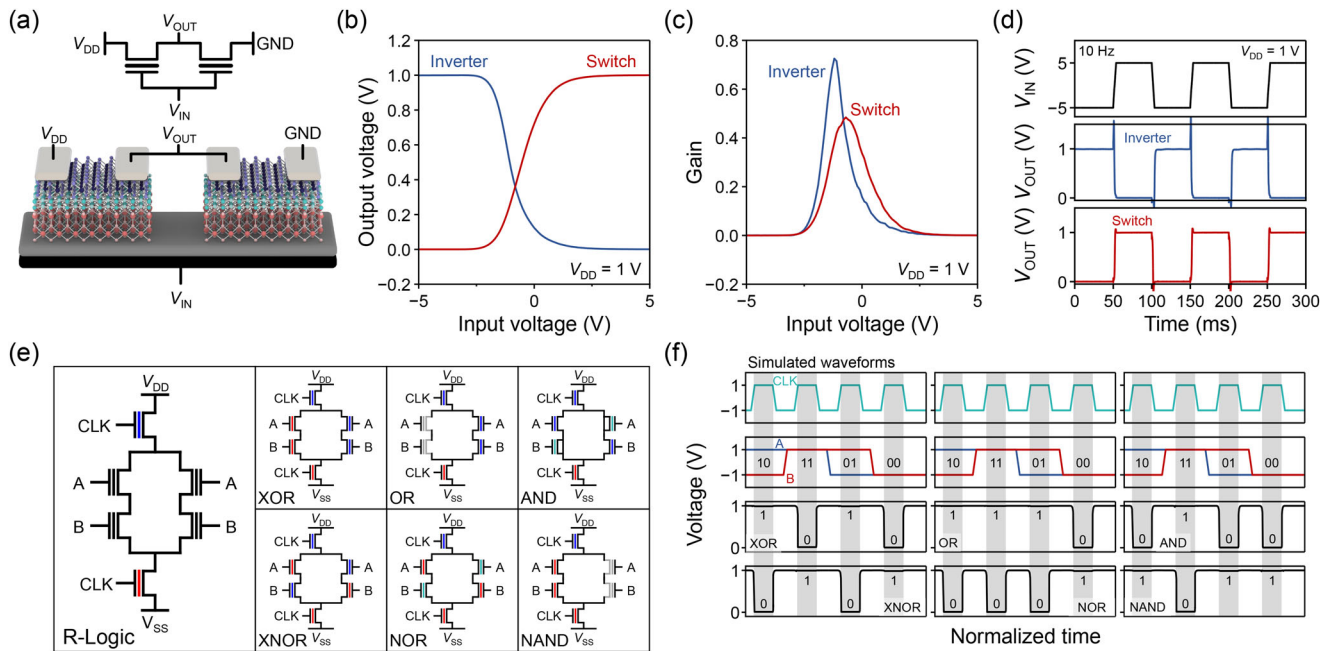


FIGURE 3 | Reconfigurable logic operation enabled by the FG mode of the hybrid-gate R-FET. (a) Circuit and measurement schematics of reconfigurable inverter and switch circuits based on hybrid-gate R-FETs. (b) Voltage transfer characteristics of the reconfigurable inverter and switch circuits. (c) Voltage gain characteristics of the reconfigurable inverter and switch circuits. (d) Dynamic response characteristics of the reconfigurable inverter and switch circuits. (e) Programming configurations for implementing various reconfigurable logic gates using the four programmable conductance states of the hybrid-gate R-FET. (f) Simulated input and output waveforms demonstrating the operation of various reconfigurable logic gates.

in series on a shared $\text{MoTe}_2/\text{h-BN}/\text{In}_2\text{Se}_3$ heterostructure and share a common bottom gate V_{IN} . The two transistors can be independently programmed into different polarity states while being driven by the same input signal. For inverter operation, the transistor connected to V_{DD} is programmed as a p-FET, while the one connected to GND is programmed as an n-FET, forming a CMOS-like configuration. In contrast, for switch operation, the programmed polarities are reversed, with the V_{DD} -side transistor as an n-FET and the GND-side as a p-FET. Figure 3b presents the corresponding voltage transfer characteristics measured at $V_{DD} = 1$ V. The inverter mode exhibits typical voltage transfer behavior, with V_{OUT} decreasing from 1 to 0 V as V_{IN} sweeps from -5 to $+5$ V. In contrast, the switch mode exhibits the opposite trend, with V_{OUT} rising from 0 to 1 V over the same input range. This response preserves the input polarity at the output, thereby acting as a non-inverting buffer within the same circuit layout.

To further characterize the operational performance, we evaluated the voltage gain and dynamic response of the two configurations. The differential voltage gain ($|dV_{OUT}/dV_{IN}|$) extracted from the V_{OUT} - V_{IN} curves is shown in Figure 3c. At $V_{DD} = 1$ V, the inverter mode reaches a peak gain of 0.72, whereas the switch mode reaches 0.48, with both peaks located near $V_{IN} = 0$ V. The higher inverter gain is attributed to the steeper transition in its V_{OUT} - V_{IN} response, which originates from the complementary push-pull operation of the two oppositely polarized transistors. Figure 3d presents the dynamic switching characteristics measured under a 10 Hz square-wave input alternating between -5 and $+5$ V at $V_{DD} = 1$ V. In the inverter mode, V_{OUT} toggles between 1 V and 0 V with a 180° phase relationship to the input, showing clear logical inversion in the temporal domain. In contrast, the

switch mode produces an output that follows the input in phase, consistent with its non-inverting transfer characteristics. The rise and fall times, extracted at the 10% and 90% levels of the output amplitude, are 1.12 ms for the inverter mode and 1.22 ms for the switch mode (Figure S7). Together, these results demonstrate stable and reconfigurable inverter-switch operation within a single device layout.

Building on this dual-mode reconfigurability, we further extended the design to a Boolean logic architecture capable of implementing six fundamental two-input gates within a single circuit platform (Figure 3e). In the initial state, the transistor connected to V_{DD} is programmed as a p-FET (blue), while the one connected to V_{SS} is programmed as an n-FET (red). Both transistors are controlled by a clock signal (CLK). The four input-driven transistors at the center of the circuit can be selectively programmed into one of the four conductance states established in Figure 2. The available states are p-FET (blue), n-FET (red), insulating (gray), and metallic (cyan). By assigning the four states across the central transistors, the same layout reconfigures into XOR, OR, AND, XNOR, NOR, and NAND gates without any modification of interconnects (Figure 3e, right). To evaluate the functional correctness of the reconfigured gates, we performed circuit-level simulations in the Cadence Spectre environment using a Verilog-A-based lookup table (Figure 3f). The lookup table was calibrated to the experimental device characteristics in Figure 2. The CLK signal was applied as a pulse alternating between -1 and $+1$ V with a pulse width of 50 ms, and the outputs were sampled at $\text{CLK} = +1$ V. The four input combinations of $(A, B) = (1, 0), (1, 1), (0, 1),$ and $(0, 0)$ were sequentially applied in synchronization with the CLK signal. The simulated XOR,

OR, and AND outputs reproduce the expected Boolean truth tables, producing sequences of (1, 0, 1, 0), (1, 1, 1, 0), and (0, 1, 0, 0), respectively. Their complementary XNOR, NOR, and NAND counterparts simultaneously produce the inverted sequences (0, 1, 0, 1), (0, 0, 0, 1), and (1, 0, 1, 1), all within the same circuit layout. Logic “1” and “0” correspond to V_{DD} (1 V) and GND (0 V), respectively. These results highlight a key distinction from previous In_2Se_3 -based reconfigurable logic studies that primarily relied on ferroelectric polarization for logic-state control [40, 41]. The stronger doping capability of the FG mode, combined with the ambipolar transport of MoTe_2 , enables the hybrid-gate R-FET to realize six representative Boolean logic functions within a single fixed circuit layout without physical modification of the circuit topology. Moreover, the same devices additionally operate as analog synapses in the FE mode, as described in the next section, so that reconfigurable logic and synaptic operation are combined within a single device platform (as summarized in Table S3).

2.4 | Synaptic Operation in the FE Mode of the Hybrid-Gate R-FET

In addition to the FG mode, the hybrid-gate R-FET can operate as a three-terminal synaptic transistor by exploiting the ferroelectric nature of In_2Se_3 . Figure 4a illustrates the FE-mode operation of the hybrid-gate R-FET. In the FE mode, synaptic weights are modulated by applying voltage pulses to V_{WC} while a small drain voltage is used for channel-current readout to minimize disturbance of the memory state. The gate voltage generates an electric field primarily along the out-of-plane direction of the In_2Se_3 layer, thereby gradually aligning its out-of-plane ferroelectric polarization. The resulting polarization-induced electrostatic field couples to the MoTe_2 channel across the h -BN layer and gradually modulates its carrier density and conductance, enabling analog synaptic weight updates. This gradual conductance modulation enables low-power analog weight tuning required for synaptic operation, in contrast to the larger and more abrupt conductance changes associated with the FG mode. To investigate the out-of-plane ferroelectricity of In_2Se_3 , PFM measurements were performed using the measurement configuration illustrated in Figure S8. Figure 4b presents the box-in-box PFM phase mapping, obtained by applying -10 and $+10$ V to the inner and outer square regions. The mapping shows clear square-domain contrast with well-defined boundaries, indicating polarization switching in the In_2Se_3 layer. Furthermore, the 180° PFM phase reversal observed in Figure 4c, together with the butterfly-shaped amplitude hysteresis presented in the Figure S8, further confirms the out-of-plane ferroelectric nature of the In_2Se_3 layer [42, 43], supporting the FE mode operation of the hybrid-gate R-FET.

Figure 4d illustrates the mechanism of synaptic potentiation and depression in the hybrid-gate R-FET. Before synaptic operation, the device is programmed into the p-FET state through the FG mode. The p-FET state is chosen because it shows a smaller subthreshold swing and a higher on/off ratio than the n-FET state. When a negative excitatory spike is applied to V_{WC} , the ferroelectric polarization in In_2Se_3 aligns toward the SiO_2 layer. This polarization induces additional hole accumulation in the MoTe_2 channel, thereby raising the channel conductance and

producing synaptic potentiation. When a positive inhibitory spike is applied, the polarization reverses and reduces the hole concentration, thereby lowering the channel conductance and producing synaptic depression. Figure 4e presents the postsynaptic current (PSC) measured before and after excitatory and inhibitory spikes. Potentiation was induced by -10 V pulses and depression by $+10$ V pulses, both 10 ms wide and applied to V_{WC} at $V_{DS} = 0.1$ V. After each pulse, the channel current shifts to a new level and remains there, giving the memory behavior required for synaptic operation. Beyond retention, the FE mode also offers a substantial advantage over the FG mode in terms of programming conditions. The FG mode requires programming pulses of ± 50 V with pulse widths of 1–1.5 s, whereas the FE mode operates using only ± 10 V pulses with a pulse width of 10 ms. As summarized in Table S4, conventional FG-based synaptic devices generally require more demanding programming conditions, while reducing the programming voltage typically relies on additional optimization of the gate stack. In contrast, by selectively utilizing the FE mode for synaptic weight updates instead of relying solely on FG operation, the proposed hybrid-gate architecture effectively relaxes the programming requirements while preserving the wide-range conductivity modulation of the FG mode. This selective operation makes the hybrid-gate R-FET a promising platform for low-power and energy-efficient neuromorphic computing. To assess the analog synaptic response, we examined the multilevel conductance modulation under repeated pulses (Figure 4f). Potentiation and depression were driven by incremental step pulses from -2.5 to -10 V and from $+2.5$ to $+10$ V, each 10 ms wide at 8 Hz. Over 64 potentiation and 64 depression pulses, the conductance spans a dynamic range of about 100, with potentiation and depression linearities of 3.69 and 3.725. These values correspond to 6-bit synaptic states, sufficient for accurate weight updates in a spiking neural network.

To evaluate the feasibility of a spiking neural network (SNN), we performed hardware-level simulations based on the conductance modulation of Figure 4f. The simulations used the MATLAB framework from our previous work [44] and targeted the MNIST handwritten-digit dataset. Figure 4g depicts the network architecture and its circuit diagram. During training, the 28×28 image pixels are mapped to input neurons, and larger pixel intensities generate longer presynaptic spike timings t_{pre} (Figure S9). The V_{pre} spikes drive the gate terminals of the FE mode synaptic devices, producing a weighted current set by the programmed conductance. A leaky integrate-and-fire (LIF) circuit integrates these currents and fires an output spike V_{post} when the output exceeds the threshold V_T . The V_{pre} and V_{post} spikes then update the synaptic weights through a simplified spike-timing-dependent plasticity (STDP) rule (Figure S9). During inference, the trained conductance act as synaptic weights that scale the input spikes into weighted output currents, and the input digit is assigned to the output neuron with the strongest firing response (winner-take-all). Figure 4h shows the recognition as a function of the number of training samples for a network with 400 output neurons. The accuracy rises with the number of training samples and reaches about 90%. Figure 4i presents the confusion matrix for 400 neurons, where strong diagonal components indicate close agreement between inferred and target digits. These results show that the FE mode of the hybrid-gate R-FET can support energy-efficient synaptic operation and SNN-based neuromorphic computing.

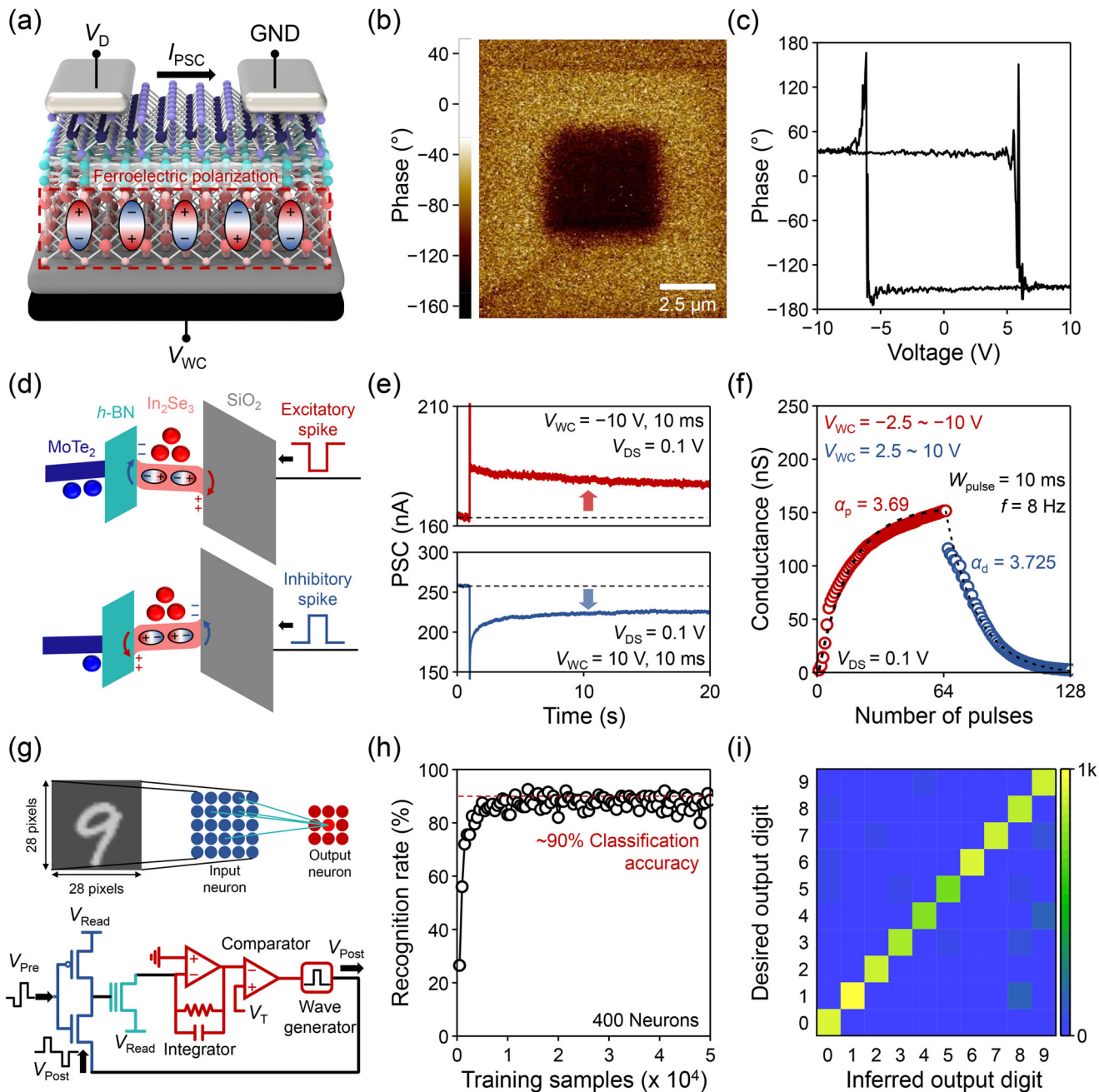


FIGURE 4 | FE mode operation and synaptic applications of the hybrid-gate R-FET. (a) Schematic illustration of the hybrid-gate R-FET operating as a synaptic transistor in the FE mode. (b) PFM phase mapping of the In_2Se_3 layer after box-in-box poling. (c) PFM phase hysteresis loop measured from In_2Se_3 layer. (d) Energy band diagrams illustrating the synaptic operation mechanism of the hybrid-gate R-FET in the FE mode. (e) PSC responses under excitatory and inhibitory gate-voltage pulses. (f) Multi-level conductance modulation exhibiting 6-bit synaptic weight states in the FE mode. (g) Schematic of the hardware-level SNN architecture and the corresponding neuron circuit used for the simulation. (h) Simulated MNIST classification accuracy as a function of the number of training samples for a network with 400 output neurons. (i) Simulated confusion matrix obtained from MNIST classification using the trained network.

3 | Conclusion

In this study, we demonstrated a hybrid-gate R-FET built on a single van der Waals gate stack that unifies FG and FE operation in one device. The two modes originate from the combined semiconducting and ferroelectric properties of In_2Se_3 , selected by the programming conditions. In the FG mode, charge stored

in the In_2Se_3 layer strongly dopes the ambipolar $MoTe_2$ channel, giving stable reconfiguration among insulating, metallic, n-FET, and p-FET states. On this basis, we experimentally realized reconfigurable inverter and switch operation, and through circuit-level simulation extended the same layout to six Boolean logic gates. In the FE mode, polarization switching of In_2Se_3 modulates the channel conductance at low voltage and low energy. The device

showed stable synaptic potentiation and depression, together with 6-bit multilevel conductance states over a dynamic range of about 100. A hardware-level spiking neural network, built on the measured synaptic characteristics, achieved about 90% recognition accuracy on the MNIST dataset through MATLAB-based simulation. By assigning logic and synaptic tasks to the FG and FE modes, the hybrid-gate R-FET overcomes the trade-offs of conventional single-mechanism R-FETs. This design establishes a single reconfigurable platform for both logic and neuromorphic computing.

4 | Experimental Section/Methods

4.1 | Fabrication of the vdW Hybrid-Gate R-FET Device

The vdW hybrid-gate R-FET was fabricated on a heavily p-type-doped Si substrate with a 90-nm-thick thermally grown SiO₂ layer. Mechanically exfoliated MoTe₂, *h*-BN, and In₂Se₃ flakes (purchased from 2D Semiconductors) were sequentially stacked to form the vdW heterostructure. The exfoliation and transfer processes were carried out using a home-built transfer stage combined with a polydimethylsiloxane (PDMS)-assisted dry-transfer method. Source and drain electrodes were patterned using a conventional photolithography process with AZ5214 photoresist. Subsequently, Pt (20 nm)/Ag (200 nm) source and drain electrodes were deposited by electron-beam evaporation and defined by a lift-off process.

4.2 | Structural and Ferroelectric Characterization of the Hybrid-Gate R-FET

For structural and compositional characterization, a cross-sectional lamella of the MoTe₂/*h*-BN/In₂Se₃/SiO₂ heterostructure was prepared using a focused ion beam (FIB) system (NX5000, Hitachi). The microstructural and elemental distributions of the heterostructure were subsequently analyzed using a transmission electron microscope (Tecnai G2 F30 S-Twin, FEI) operated at an acceleration voltage of 300 kV in conjunction with energy-dispersive x-ray spectroscopy (EDS). For piezoresponse force microscopy (PFM) measurements, mechanically exfoliated In₂Se₃ flakes were transferred onto a Si substrate with a 90-nm-thick thermally grown SiO₂ layer. The ferroelectric properties of the In₂Se₃ flakes were characterized using an NX100 atomic force microscope (Park Systems) operating in PFM mode.

4.3 | Electrical and Synaptic Characterization of the Device

Electrical characterization of the device, including transfer characteristics ($I_{DS}-V_{GS}$) and inverter voltage transfer characteristics, was performed using a semiconductor parameter analyzer (B1500A, Keysight). Synaptic and pulse-response measurements were conducted by monitoring the time-dependent electrical signals using a source/measure unit (B2912B, Keysight), while the electrical pulse stimuli were generated using a waveform generator (33500B, Keysight). All electrical bias conditions used in this study are provided in the corresponding figures.

4.4 | Circuit and SNN Simulations

For circuit simulations, the device was modeled using a Verilog-A-based lookup table (LUT) approach. The channel resistance as a function of gate voltage was extracted from the experimentally measured transfer characteristics and implemented in the LUT model to reproduce the device behavior. All circuit simulations were performed using Cadence Spectre. Hardware-level spiking neural network (SNN) simulations were conducted in a MATLAB environment based on a previously reported framework [38]. The experimentally obtained synaptic characteristics were modeled using the fitting equations provided in the Supporting Information and incorporated into the simulation framework. The classification performance was evaluated using the MNIST handwritten digit dataset.

Author Contributions

H.S., S.L., and D.-H.K. designed the experiments and analyzed the data. S.-U.A. conducted PFM analysis for ferroelectric materials. H.H.J., Y.N.A., G.Y., D.N., H.-J.S., and H.H.Y. fabricated hybrid-gate R-FET devices. H.S. and S.L. conducted TEM analysis. H.S., S.L. and I.-M.Y. designed circuit and SNN simulation framework and conducted the simulations. H.S., S.L., and D.-H.K. prepared the manuscript. All authors revised the manuscript. D.-H.K. guided the entire project.

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Conflicts of Interest

The authors declare no conflicts of interest.

Data Availability Statement

The data that support the findings of this study are available from the corresponding author upon reasonable request.

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